

ATDM & Physics

Hooke's Law

Objects

- Spring S
 - Change with respect to resting length ("change in length of S ", $\Delta\vec{x}$) $\in \text{String} \times \mathbb{R}^3$
 - Spring constant ("spring constant of S ", k) $\in \text{String} \times \mathbb{R}$
 - Force ("force on S ", $\vec{F}_S$) $\in \text{String} \times \mathbb{R}^3$
- Weight E (some heavy object dangling from the spring, assume spring has no mass)
 - Weight ("weight of E ", $\vec{W}$) $\in \text{String} \times \mathbb{R}^3$
 - Force ("force on E ", $\vec{F}_E$) $\in \text{String} \times \mathbb{R}^3$
- Nature: Union of S and E

Actions

- Application of weight
 - So it's trivially a function $E \rightarrow S$.

$$\vec{F}_S = \vec{W}$$

Responses

- Action-reaction
 - Again, the trivial negation function $S \rightarrow E$.

$$\vec{F}_E = -\vec{F}_S$$

Behavior of S

- Change in length caused by force
 - Hooke's law is the following

$$\vec{F}_S = -k\Delta\vec{x}$$

We want to calculate $\Delta\vec{x} = -\frac{\vec{F}}{k}$. This is effectively a function $S \times S \rightarrow S$, hence it's $\oplus S \rightarrow S$.

- Problem: Some behaviors involve more than three properties. Structure is a temporary hack for two inputs. So, is there any better way to do Hooke's Law?

Thermodynamics

Objects

- Body S
 - Mass ("mass of S ", m_S) $\in \text{String} \times \mathbb{R}$
 - Change in temperature ("change in temperature of S ", ΔT_S) $\in \text{String} \times \mathbb{R}$
 - Specific heat ("specific heat of S ", c_S) $\in \text{String} \times \mathbb{R}$
 - Change in thermal energy ("change in thermal energy of S ", ΔQ_S) $\in \text{String} \times \mathbb{R}$
- Environment E (probably air)
 - Change in thermal energy ("change in thermal energy of E ", ΔQ_E) $\in \text{String} \times \mathbb{R}$
- Nature: Union of S and E

Actions

- Heat
 - Heat transfer is antisymmetric.

$$\Delta Q_S = -\Delta Q_E$$

Responses

- Heat
 - Same equation.

$$\Delta Q_E = -\Delta Q_S$$

Behavior of S

- Temperature change caused by transfer of heat.

$$\Delta Q_S = c_S m_S \Delta T_S$$

$$\Rightarrow \Delta T_S = \frac{\Delta Q_S}{c_S m_S}$$

- Problem: this illustrates the problem previously mentioned about more than two inputs. Here, it's a function $S \times S \times S \rightarrow S$.

First law of thermodynamics

- The first law is conservation of energy. This is effectively illustrated when considering the antisymmetry of the action and response formulas.

Newtonian Gravity

Objects

- Particle O_X (primitive)
 - Mass ("mass of X", m) $\in \text{String} \times \mathbb{R}$
 - Position ("position of X", $\vec{x}$) $\in \text{String} \times \mathbb{R}^3$
 - Velocity ("velocity of X", $\vec{v}$) $\in \text{String} \times \mathbb{R}^3$
 - Acceleration ("acceleration of X", $\vec{a}$) $\in \text{String} \times \mathbb{R}^3$
 - Proper time ("proper time of X", τ) $\in \text{String} \times \mathbb{R}$
 - *Note*: Position and velocity are relative to a reference point. The rest of the notes will deal with the issues it raises.
- Nature: Union of all particles

Actions

- Gravity
 - Consider the general gravity formula for O_A attracting O_B . We treat $O_B = S$ as the product, thus $O_A \subseteq E$ belongs to the environment.

$$\vec{F}_{A,B} = G \frac{m_A m_B}{\|\vec{x}_A - \vec{x}_B\|^3} (\vec{x}_A - \vec{x}_B)$$

- If we let O_B be a reference point, i.e. S is the observer, then $\vec{a}_B$ is computed as

$$\vec{a}_B = \frac{\vec{F}_{A,B}}{m_B} = G \frac{m_A}{\|\vec{x}_A - \vec{x}_B\|^3} (\vec{x}_A - \vec{x}_B) = G \frac{m_A}{\|\vec{x}_A\|^3} \vec{x}_A$$

where $\vec{x}_A$ is now the relative position of O_A to O_B . This is the case for a single-particle environment. For a multi-particle environment, simply compute the sum for every particle in the environment $O_\alpha \subseteq E$.

- As seen from the formula, the computation of $\vec{a}_B$ involves only $O_a \subseteq E$, and maps to the acceleration of $O_B = S$, hence it has the form

$$f : E \rightarrow S$$

Thus, the acceleration caused by the environment is an action, i.e. $f \subseteq E \times S$.

- Problem: Given any reference frame, the position/velocity/acceleration of the observer is always zero. In other words, we are likely referring to *regular* acceleration $\vec{a}_B$, which is dependent on its relative motion with the other particles. The next section contains a candidate solution.

Responses

- Gravity

- The particle $O_B = S$ also creates acceleration for other $O_\alpha \subseteq E$.

$$\vec{a}_\alpha = -G \frac{m_B}{\|\vec{x}_\alpha\|^3} \vec{x}_\alpha$$

- Problem: If we keep the same reference frame, then this is a function $g : (E \times S) \rightarrow E$. One way to resolve this issue is to store, within each particle, its relative position as seen from every particle (including itself, which is the trivial case, which generates $(0, 0, 0)$). This is equivalent to smashing together the position content of $E \times S$ into S , thus we can simply have a map $S \rightarrow E$. In the context of the previous problem, we do the corresponding pairwise storage, but for acceleration.